

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc (Hardware And Networking)
as per NEP 2020

w.e.f session 2025-26

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-101	Core Discipline	Basics of Computer System	4	--	4	80	20	--	--	100
BHN-102	Core Discipline	Fundamental of PC Hardware	2	--	4	40	10	--	--	100
BHN-102P		Practical Based on BHN-102	--	2		--	--	40	10	
BHN-103	Core Discipline	PC Configuration	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva- Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-201	Core Discipline	Data Communication and Networks	4	--	4	80	20	--	--	100
BHN-202	Core Discipline	Computing Software	2	--	4	40	10	--	--	100
BHN-202P		Practical Based on BHN-202	--	2		--	--	40	10	
BHN-203	Core Discipline	Network Configuration	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	--	4	4	--	--	80	20	100
Total Credits					**24**					**600**

Job Role - Field Technician- Computing & Peripherals

*Summer Industrial Training of 2-4weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3-weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in (HARDWARE AND NETWORKING) =48 + 4 #.
#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

***One credit of theory lecture is of 1 hour duration and one credit of practical is of 2 hours duration.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-301	Core Discipline	Basics of Routing and Switching	4	--	4	80	20	--	--	100
BHN-302	Core Discipline	Programming using C Language	2	--	4	40	10	--	--	100
BHN-302P		Practical Based on BHN-302	--	2		--	--	40	10	
BHN-303	Core Discipline	Routing Configuration	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 301	Multidisciplinary Course	Entrepreneurship Development[1]	3	-	3	60	15	-	-	75
SEC III	Skill Enhancement Course	SEC Options[2]			3					75
VAC II	Value-Added Course	VAC Options[3]			2					50
PW II	Project Work	Project Work-II & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
BHN-401	Core Discipline	Digital Electronics and Computer Organisation	4	--	4	80	20	--	--	100
BHN-402	Core Discipline	Network Administration	2	--	4	40	10	--	--	100
BHN-402P		Practical Based on BHN-402	--	2		--	--	40	10	
BHN-403	Core Discipline	DOS and Windows Operating System	2	--	4	40	10	--	--	100
BHN-403P		Practical Based on BHN-403	--	2		--	--	40	10	
BHN-404	Core Discipline	Switching Configuration	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
VAC III	Value-Added Course	VAC Options[3]			2					50
SIT 401	Internship	Summer Industrial Training (SIT)*	--	4	4	--	--	80	20	100
Total Credits					**24**					**600**

Job Role - Network Support Engineer

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 5th Semester. Viva-Voce examination to be held within 3-weeks of the start of 5th semester.

** Minimum Credits Required to get Diploma Course in (HARDWARE AND NETWORKING) =96 + 4 #.
#Exit 2: Diploma Course-Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the second year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

***One credit of theory lecture is of 1 hour duration and one credit of practical is of 2 hours duration.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF **B.Voc (HARDWARE AND NETWORKING)** (SEMESTERS I & II)

PROGRAMME OUTCOMES

1. The primary objective of this course is to prepare students for careers as Field Technicians (Computing and Peripherals) and Hardware and Network Engineers/Administrators. These roles are crucial in the IT industry, particularly in servicing firms such as telecom companies and equipment manufacturing firms. Students will gain the necessary skills and knowledge to excel in these positions.

2. The course places a strong emphasis on practical, hands-on training. It recognizes the importance of providing students with an early platform to manage and troubleshoot complex failures in computer hardware and networks. By engaging in practical exercises, students will develop the skills required to analyze problems and implement effective solutions, fostering a deeper understanding of the field.

3. A key focus of the curriculum is to cultivate the ability of students to identify and resolve issues related to computer hardware and networks. The course structure is designed to equip students with problem-solving skills, enabling them to troubleshoot and address challenges in a real-world context. This emphasis on practical problem-solving enhances the employability of graduates.

4. The curriculum is well-rounded, encompassing both general education and skill development components. Beyond technical expertise, students will receive a broad education that includes relevant industry knowledge and soft skills. This comprehensive approach ensures that

graduates are not only technically proficient but also well-rounded professionals capable of adapting to diverse work environments.

5. To bridge the gap between theoretical knowledge and practical application, the course integrates extensive practical and on-the-job training components. Students will have the opportunity to apply their theoretical learning in real-world scenarios, preparing them for the challenges they may encounter in their future roles. This hands-on experience is invaluable for building confidence and competence.

6. To further connect students with the professional world, the course incorporates industry interactions. Students will have the chance to engage with professionals in the field, participate in workshops, and attend seminars. This exposure facilitates networking opportunities, allowing students to learn from industry experts, stay updated on industry trends, and potentially secure internships or job placements. This connection to the industry is instrumental in preparing students for a successful transition from education to the workforce.

Semester I

<table>
<tr><th colspan="2">Course Title</th><th>Basics of Computer System</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-101</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
</td></tr>
</table>

		iv. The students are required to attempt one long question of 16 marks from each unit.
6	**Job Role**	Field Technician- Computing & Peripherals
7	**Course Objectives**	The course aims to equip students with a profound understanding of computer fundamentals, hardware, software, and networking. To empowers the ability to analyze advantages and disadvantages, design computer system architectures, apply input and output devices, and comprehend emerging technologies and well-rounded expertise in basics concepts.
8	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Demonstrate a comprehensive understanding of computer fundamentals, including the historical development and characteristics of computers. CO2. Apply knowledge of input and output devices, computer memory, and different types of software in practical scenarios. CO3. Analyze and evaluate the functions and types of operating systems, data processing systems, and translators. CO4. Demonstrate proficiency in understanding computer networks, the Internet, and emerging technologies, with the ability to classify and describe different types of networks.
9	**Pedagogy of the Course Work**	Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.
10	**Course Content**	

UNIT– I

Computers: Characteristics, Historical development of computers, Evolution of computing devices, Generations of Computers: First Generation (Vacuum Tubes, Batch Processing), Second Generation (Transistors), Third Generation (Integrated Circuits, Time-Sharing Systems), Fourth Generation (Microprocessors, Personal computers), Fifth Generation (Artificial intelligence). Hardware, Software and Firmware, Data and Information. Computer Codes: Decimal System, Binary System, Hexadecimal System, Octal System and ASCII Code. Decimal to Binary conversion, Binary to Decimal Conversion.

UNIT– II

Introduction to Computer Networks: Definition and Importance of Computer Networks, Basic Components of a Computer Network, Classification of Networks, Network Topologies. Internet: Historical Overview of The Internet, Evolution of The World Wide Web (WWW), WWW Components, Internet Governance and Standards, Internet Services and Applications (Email Protocols, Telnet, FTP, IRC, Usenet News). Internet Access Methods (Dial-Up, DSL, Cable, Fiber-Optic), Browser Technologies and Features.

UNIT– III

Computer Virus: Types of Viruses, Impact of Virus, Virus Detection and Prevention techniques, Malware. Antivirus and Its Types. Multimedia: Hardware and Software for Multimedia, Multimedia: Component of Multimedia, Basic File Formats, Audio File Formats, Image File Formats, Video File Formats. Multimedia Storage Devices.

UNIT-IV

Emerging Technologies: Basics of Internet of Things (IoT), Components of IoT, Applications of IoT, Basics of Parallel Computing and Distributed Computing, Difference between Parallel computing and Distributed Computing, Advantages and Limitations of Parallel computing and Distributed Computing, Cloud Computing: Component of Cloud Computing, Types of Cloud Computing, and their applications.

11	**Suggested Resources for Reading** 1. Fundamentals of Computers, V. Rajaraman, PHI Publication. 2. Computer Fundamentals, P. K. Sinha, BPB Publication. 3. Computers Today, Basandra, S. K., Galgotia Publications. 4. Fundamentals of Information Technology, Anshuman Sharma, Lakhanpal Publications. 5. PC Hardware: The Complete Reference, Craig Zacker., TMH Publication. **Note:** Latest edition of text books may be used.

Course Title		**Fundamental of PC Hardware**
1	**Course Code**	**BHN-102**
2	**Typology of Course**	Core Discipline
3	**Course Duration**	30 hours of lectures for Theory + 60 hours of practicals Theory credits =2 and Practical credits=2 (1 credit = 15 hours of theory learning based on 2 hrs./week) + (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 2-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits
5	**Max. Marks, Time and Instructions**	**External Theory**: 40; **Internal Theory**: 10; **External Practical**: 40; **Internal Practical**: 10 **Theory Paper:** **(4X 2 marks) = 8 marks** **(4X8 marks) = 32 marks** i. The exam of this course will be of 3 hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees. **Instructions for paper setters and candidates:** i. The syllabus of this paper has been divided into FOUR units. ii. All questions carry equal marks iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each. iv. The students are required to attempt one long question of 8 marks from each unit.
6	**Job Role**	Field Technician- Computing & Peripherals
7	**Course Objectives**	This course aims to provide comprehensive knowledge of computer peripherals, hardware components, and system management. Hardware Fundamentals enable to learn about hardware technologies, their implementation/installation, uses, troubleshooting and maintenance.

8	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Demonstrate a comprehensive understanding of computer hardware components, peripherals, and their functionalities. CO2. Enable the students to learn about hardware technologies and their implementation/installation, uses, troubleshooting and maintenance. CO3. Install, update, and troubleshoot drivers for various hardware components, including network adapters, sound cards, and graphics cards. CO4. Configure system settings, understand the role of BIOS, and troubleshoot common BIOS-related issues.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.
10	**Course Content** **UNIT-I** Peripheral Devices: Introduction to Peripheral Devices, Functionality, Use and Types of Various Peripheral Devices (Mouse, Keyboard, Printer, and Monitor). Add-On Cards and Its Functionality: Network Interface Card, Sound Card and Graphics Card. Uninterruptible Power Supply (UPS): Features, Types (Offline & Online UPS), Advantages & Limitations of UPS. Mother Board: Definition, Importance, Basic Components, Layout of Motherboard, Types of Motherboards (AT, ATX, and BTX). **UNIT-II** Ports: Definition, Role of Ports in Computer Systems, Functions, Types of Ports: Serial Ports and its Types, Parallel Ports, and its Types. Hard Disk: Definition, Importance of Hard Disk, Structure of Hard Disk (Head, Platter, Spindle, Actuator, Tracks, Sectors), Types of Hard Disk (PATA, SATA, SCSI). RAM: Definition, Uses, Types of RAM (SRAM, DRAM, SDRAM). Power Supply Unit: Functions, Components, Power Connectors, Power Requirements for various Computer Components. Processor: Definition, Architecture, Functions, Clock Speed, Cores, and Cache Memory. **UNIT-III** Cables: Importance of Cables in Computer Systems, Types (VGA, DVI, RJ11, RJ45, HDMI, Serial, Parallel Cables, USB (Versions, Specifications, Common Uses, Connectors and Compatibility). Computer Cable Management: Importance of Cable Management for Airflow and Aesthetics, Techniques for Organizing and Securing Cables, Computer Case Management: Types, Form Factors, Importance of Airflow and Ventilation in Computer Cases, Reducing Cable Clutter Inside the Computer Case. Drivers: Definition, Role of Device Drivers, Importance of Drivers for Hardware Functionality, Find and Install Device Drivers. Types of Drivers (Device Drivers, LAN Drivers, Sound Drivers, Graphics Drivers). Cabinet: Role of Cabinet, Types of Computer Cabinets. **UNIT-IV** BIOS: Definition, Importance, Functions, Responsibilities, Role of BIOS in the Boot Process, Importance of BIOS Updates, Procedure for Updating BIOS, Risks and Precautions during BIOS Updates, Common BIOS Error Messages. Antivirus Software: Definition, Features, Significance, Role of Antivirus in Security, Installing and Configuring any Third-Party Antivirus for Optimal Protection. Firewall: Definition, Purpose of Firewalls, Importance in Network Security, Hardware Vs. Software Firewalls, Types of Firewalls (Packet-Filtering Firewalls, Stateful Inspection Firewalls, Proxy Firewalls and Application-Layer Filtering), Limitations and Challenges in Firewall Protection, Real World Examples of Successful Firewall Applications.	
11	**Suggested Resources for Reading** 1. The Indispensable PC Hardware Book (4th Edition), Hans-Peter Messmer, Addison Wesley. 2. PC Hardware: A Beginner's Guide by Ron Gilster, McGraw-Hill Education Note: Latest edition of text books may be used. 3. PC Hardware: The Complete Reference, Craig Zacker., TMH Publication. 4. Mastering PC Hardware & Network, Dr. Ajit Mittal, Dr. Ajay Rana, Khanna Book Publishing Company	

	5. How Computers Work, Ron White and Timothy Edward Downs, QUE. 6. Fundamentals of Computers, V. Rajaraman, PHI Publication. **Note**: Latest edition of text books may be used.
BHN-102P (Practical Based on BHN-102) Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation. 1. Examiner will set total of four questions covering the whole syllabus. 2. Student will attempt any three questions. 3. Each question carries 5 marks. 4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.	

Course Title		**PC Configuration**
1	**Course Code**	**BHN-103**
2	**Typology of Course**	Core Discipline
3	**Course Duration**	120 hours of practicals Practical credits=4 (1 credit = 30 hours of practical based on 8 hrs./week) (1Semester=15weeks) 2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits
5	**Max. Marks, Time and Instructions**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>-</td><td>-</td><td>80</td><td>20</td></tr></table> i. The exam of this course will be of 3hours duration. ii. There will be no objective type questions. iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees **Instructions for paper setters and candidates:** i. Examiner will set total of four questions covering the whole syllabus. ii. Student will attempt any three questions. iii. Each question carries 10 marks. iv. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
6	**Job Role**	Field Technician- Computing & Peripherals
7	**Course Objectives**	The course aims to provide a comprehensive understanding of computer systems, covering fundamental hardware components. It develops practical skills in connecting peripherals, analyze and solve hardware issues logically, and assemble, configure, troubleshoot, and maintain functional computer systems. Additionally, the course focuses on installing Windows, managing drivers, and implementing antivirus software for system security.

8	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Define and explain the fundamental components of a computer system, demonstrating a holistic understanding of hardware architecture. CO2. Demonstrate practical skills in connecting and configuring peripherals, including mouse, keyboard, display devices, scanning devices, and hard disk. CO3. Develop critical thinking skills to analyze and solve hardware-related problems, applying a logical and systematic approach to diagnose and address issues. CO4. Assemble, configure, troubleshoot, and maintain a functional computer system, integrating knowledge from all units for practical application. CO5. Successfully install the Windows operating system, manage the installation and updating of drivers, install and update anti-virus software.
9	**Pedagogy of the Course Work**	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
10	**Course Content**	

UNIT-I

Introduction To Computer Hardware:

- Overview of Computer Components
- Basic Understanding of CPU, RAM, Storage Devices, Various Ports, Slots (Add On Devices) and Input/Output Devices (Keyboard, Mouse, Monitor, Printer).
- Hands-On Practice Connecting and Configuring Peripherals (Mouse, Keyboard, Display Devices, Scanning Devices, Hard Disk).
- Studying Various Components of The Motherboard.
- Introduction to Hardware Tools and Safety Precautions

UNIT-II

Computer Cables:

- Various Types of Cables Used in Computer, Connectors and Ports used.
- Connecting Cables from SMPS To Motherboard & Hard Disk.
- Establishing Data Connection for Mother Board, Hard Disk, and Drivers.
- Fixing Wires for Power Restart, Fixing Wires for HDD, LED, External USB & Audio Connections.
- Cable Management Techniques.

UNIT-III

System Diagnostics, Troubleshooting and Maintenance:

- Overview of the Troubleshooting Process and the Importance of Systematic Approach
- Introduction To Hardware Diagnostic Tools (PC Toolkit)
- Troubleshooting Common Hardware Problems in Keyboard, Mouse, Mouse, Printer, SMPS, Hard Disk, Motherboard, and Processor.
- Maintenance of Keyboard, Mouse, Mouse, Printer, SMPS, Hard Disk, Motherboard, and Processor.
- Breakdown and Breakthrough of various Hardware Issues.
- Safety Precautions During Troubleshooting.

UNIT-IV

Window Installation:

- Installing Drivers.
- Installing and Updating Anti-Virus Software.
- Hard Disk Partitioning (Creating, formatting & Deleting Partitions) Using DISKPART in Command Prompt.
- Hard Disk Partitioning (Creating, formatting & Deleting Partitions) using Graphical User Interface.

11	**Suggested Resources for Reading** 1. Comdex Hardware and Networking Course Kit, Gupta, V., Dreamtech Press. 2. IBM-PC and Clones – Hardware Troubleshooting and Maintenance, Govindarajalu, B., Tata McGraw Hill. 3. Mastering PC Hardware & Network, Dr. Ajit Mittal, Dr. Ajay Rana, Khanna Book Publishing Company. 4. PC Hardware: A Beginner's Guide by Ron Gilster, McGraw-Hill Education. 5. PC Hardware: The Complete Reference, Craig Zacker., TMH Publication **Note**: Latest edition of text books may be used.

Semester II

<table>
<tr><th colspan="2">Course Title</th><th>Data Communication and Networks</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-201</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
iv. The students are required to attempt one long question of 16 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Field Technician- Computing & Peripherals</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course aims to familiarize students with diverse network types, detailing their roles in computer environments. It provides a foundational understanding of network security concepts, emphasizing the importance of safeguarding infrastructure. Students will also learn switching techniques, IP address fundamentals and configuration.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Develop a comprehensive understanding of computer networks, analyze, and compare various network topologies and understand different network models.
CO2. Identify and describe different types of servers, understanding their roles and functionalities in network environments.
CO3. Understand transmission media, various networking protocols and the functionalities of network devices.
CO4. Develop a basic understanding of network security concepts, recognizing the importance of safeguarding network infrastructure.
CO5. Understanding IP addresses and its configuration.</td></tr>
</table>

9	**Pedagogy of the Course Work**	Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.
10	**Course Content** **UNIT-I** Computer Network: Introduction, Characteristics, Uses of Computer Networks, Types of Networks Based on Area Coverage: LAN, WAN, MAN, Advantages and Disadvantages of Networks, Network Topologies: Bus Topology, Ring Topology, Star Topology, Tree Topology, Mesh Topology and Hybrid Topology. Network Models: Peer-To-Peer, Protocol, Client/Server Network and Hybrid Type Network. OSI Reference Model, TCP/IP Reference Model, Comparison of OSI with TCP/IP Model. Servers: Different Types of Servers. **UNIT - II** Data Communications: Analog Signal, Digital Signal, Analog Vs Digital Communication, Band Width Limitation, Data Rate of a Channel, Physical Layer: Transmission Media: Guided (Twisted-Pair, Coaxial and Optical Fiber) and Unguided (Radio, Microwave, and Infrared), Switching: Circuit Switching, Packet Switching, Message Switching. Telephone System, Modems. Modulation Techniques: Amplitude Modulation, Frequency Modulation, Phase Modulation. Multiplexing: Frequency Division, Time Division, Wave Length Division. **UNIT-III** Networking Protocols: Introduction, Domain Name System (DNS), DNS Name Space, DNS Servers, FTP, Telnet, DHCP, HTTP, TFTP, SSH, SFTP, SMTP, POP, IMAP, NTP. Network Devices: HUB, Switch, Repeater, Bridges, Routers, Gateway. IEEE Standards: IEEE Standards for LAN and WAN: IEEE Standard 802.3 and Ethernet, IEEE Standard 802.4 Token Bus, IEEE Standard 802.5 Token Ring, IEEE Standard 802.6 Distributed Queue Dual Bus, IEEE Standard 802.2 Logical Link Control. Introduction to Network Security. **UNIT-IV** IP Address: Introduction, IP Address Format, Overview of IP Address Classes: A, B, C, D, and E, Characteristics and Range of Each IP Address Class, Use Cases for Different IP Address Classes. Subnetting-Sub Network, Subnet Mask, Configuration of IP Address in PC, Differentiation Between IPv4 and IPv6. Access Points, Network Interface Card (NIC): Working Principle of NIC, MAC Address. User Management: Overview of User Account, Managing User Account and Password Security: Adding User, Setting and Changing Password, Changing Picture of User, Deleting the User, Setting Account Restriction for The User, Renaming a Local Account.	
11	**Suggested Resources for Reading** 1. Computer Networks, Tanenbaum, Andrew, Fifth Edition, PHI. 2. Data Communication and Networking, Behrouz A. Forouzan, Fourth Edition, McGraw Hill Education. 3. Computer Today, S.K. Basandra, First Edition, Galgotia. 4. Data Communication System, Black, Ulysse, Third Edition, PHI. 5. Data and Computer Communications, Stalling, Ninth Edition, PHI. **Note:** Latest edition of text books may be used.	

<table>
<tr><th colspan="2">Course Title</th><th>Computing Software</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-202</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practicals
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Field Technician- Computing & Peripherals</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course covers essential skills in word processing, spreadsheet management, presentation creation, web browsing, and email communication. Students will learn document handling, formatting, and advanced features in Microsoft office applications. Additionally, they will acquire proficiency in web browsing, email etiquette, organization strategies, and customization options, enhancing their overall digital literacy.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Develop proficiency in using office suite applications, including word processing, spreadsheets, and presentation tools, for efficient document creation, data analysis, and presentation development.
CO2. Develop technological literacy by understanding and utilizing the latest features and tools available in office suite applications.
CO3. Students will be able to develop engaging presentations, mastering slide design, animations, and transitions, enabling effective communication through multimedia slideshows.
CO4. Understand basic communication tools, web surfing techniques, e-governance website usage, and understand email security settings, two-factor</td></tr>
</table>

		authentication, and safe practices for online communication.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.
10	**Course Content**	

UNIT-1

Word Processing Package: Opening, Saving and Closing an Existing Document, Renaming and Deleting Files, Using Styles and Templates: Introduction to Templates and Styles, Applying, Modifying and Creating New (Custom) Styles, using a Template to Create a Document, Creating a Template, Editing a Template, Organizing Templates, Examples of Style Use, Changing Document Views, Moving Quickly Through a Document.

Working With Text: Select, Cut, Copy, Paste, Find and Replace, Inserting Special Characters, Setting Tab Stops and Indents, Checking Spelling and Grammar, Autocorrect, Using Built-In Language Tools, Word Completion, Autotext, Formatting Text: Using Styles, Formatting Paragraphs, Formatting Characters, Autoformatting, Creating Lists.

Formatting Pages: Using Layout Methods, Creating Headers and Footers, Numbering Pages, Changing Page Margins, Adding Comments to a Document, creating a Table of Contents, Creating Indexes and Bibliographies, Printing a Document, Using Mail Merge, Tracking Changes to a Document, Using Fields, Linking to Another Part of a Document, Using Master Documents, Creating Fill-In Forms.

UNIT-II

Spreadsheet Package: Worksheet Basics, Data Entry in Cells: Entry of Numbers, Text and Formulae, Moving Data in a Worksheet, Moving Around in a Worksheet, Selecting Data Range, Using the Interface (Toolbars, Menus), Editing Basics, Working with Workbooks, Cell Referencing. Formatting and Calculations: Using Auto Fill, Working with Formulae, Efficient Data Display with Data Formatting (Number Formatting, Date Formatting Etc.), Working with Ranges, Worksheet Printing.

Working with Graphs and Charts: Creating Embedded Chart Using Char Wizard, Sizing and Moving Parts, Updating Charts, Changing Chart Types, Chart Wizard, Adding Titles, Legends and Gridlines, Printing Charts, Database Management. Finding Records with Data Form, Adding/Deleting Records, Filtering Records in a Worksheet. Functions and Macros: Worksheet Creating Macros, Recording Macros, Running Macros, Assigning Macros to Buttons, Defining Macros from Scratch. Multiple Worksheets.

UNIT-III

Presentation Package: Introducing Power Point, Use of Power Point Presentation, Creating New Slides Saving, Opening, and Printing. Home Tab: New Slide, Layout, Reset, Delete, Setting Text Direction, Align Text, Convert to Smart Art, Drawing Options. Insert Tab: Table, Picture, Clipart, Photo Album, Smart Art, Shapes and Chart, Movie and Sound, Hyperlink and Action, Text Box, Word Art, Object.

Design Tab: Page Setup Options, Slide Orientation, Applying Various Themes, Selecting Background Style, and Formatting it. Animations Tab: Custom Animation for Entrance, Exit and Emphasis, Applying Slide Transition, Setting Transition Speed and Sound, Animation on Rehears Timing. Slide Show &View Tab: Start Slide Show Options, Setup Options. View Tab: Presentation Views, Colours, and Window Option.

UNIT-IV

Web Browsing: Popular Web Browsing Softwares, Search Engines, Search for Content Accessing Web Browser, Using Favorites Folder, Downloading Web Pages, Printing Web Pages, Understanding URL, Surfing the Web, Using E-Governance Website.

Communication: Overview of E-Mail and its Evolution, Creating and Customizing E-Mail Account,

	Understanding E-Mail User Interface and Layout, Basic Email Etiquette, Essential Email Features, Composing, Sending, and Receiving Emails, Attachments and File Sharing, Labels and Filters for Organizing Emails, Advanced Search Techniques. Organization Strategies: Inbox Organization and Management, Priority Inbox and Tabbed Inbox Features, Using Folders (Labels) Effectively, Archiving and Deleting Emails, Advanced Settings, and Customization: Customizing E-Mail Settings for Personal Preferences, Signature Creation and Customization, Vacation Responders and Out-of-office Settings, Security Settings and Two-Factor Authentication, Mobile and offline Access: Configuring E-Mail on Mobile Devices, offline Access, and Synchronization.
11	**Suggested Resources for Reading** 1. Microsoft Windows Networking Essentials 1st Edition, Darril Gibson, Sybex. 2. Computer Fundamentals, Sinha P.K. & Sinha Priti, BPB Publications. 3. DOS 6.2 Quick Reference, Mathur Rajiv, Galgotia. 4. Introduction to Computers with MS - Office, Leon, TMH Publication 5. Using the Internet, Kasser, Barbara, PHI, 4th ed., New Delhi. 6. Using the World Wide Web, Wall, David A. & Others, PHI, 2nd ed., New Delhi. **Note**: Latest edition of text books may be used.

BHN-202P (Practical Based on BHN-202)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

Course Title		**Network Configuration**
1	**Course Code**	**BHN-203**
2	**Typology of Course**	Core Discipline
3	**Course Duration**	120 hours of practicals Practical credits=4 (1 credit = 30 hours of practical based on 8 hrs./week) (1Semester=15weeks) 2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits
5	**Max. Marks, Time and Instructions**	

External Theory	**Internal Theory**	**External Practical**	**Internal Practical**
-	-	80	20

i. The exam of this course will be of 3hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees

Instructions for paper setters and candidates:

		i. Examiner will set total of four questions covering the whole syllabus. ii. Student will attempt any three questions. iii. Each question carries 10 marks. iv. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
6	**Job Role**	Field Technician- Computing & Peripherals
7	**Course Objectives**	This course aims to provide a comprehensive understanding of networking tools, connectors, cables, and testing instruments. Participants will learn to establish LAN connections, configure internet connectivity, and implement Wi-Fi setups. The course emphasizes hands-on skills in cable termination, network testing, and troubleshooting.
8	**Course Outcomes**	**On successful completion of the course, the students will be able to** CO1. Proficiently utilize networking tools like crimping and punching tools. CO2. Demonstrate expertise in working with various connectors, including RJ-45 and Fiber LC. CO3. Construct and troubleshoot LAN connections using UTP cables, switches, and testing tools. CO4. Execute networking commands for configuration and diagnostics. CO5. Configure and secure internet and Wi-Fi connectivity, demonstrating knowledge of different internet connection types.
9	**Pedagogy of the Course Work**	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
10	**Course Content**	

UNIT-I

Working Using Various Networking Tools

- Crimping Tool, Punching Tool, Punch Down Tool, Multi Punch Tool, RJ45 Punch Down Tool, Computer Punch Tool
- Cable Cutter, I/O Device, I/O Face Plate, I/O Box, Lan Cards

Working Using Various Connectors

- RJ-11, RJ-45, F-Type, ST (Straight Tip)
- SC (Subscriber Connector or Standard Connector)
- Fiber LC (Local Connector)
- MT- RJ (Mechanical Transfer Registered Jack)
- USB (Universal Serial Bus)

UNIT-II

Networking Cables, Patch Cords and Testing Tools:

- Working of Different Cables: Coaxial Cable, Unshielded Twisted Pair (UTP), Types of UTP Cable, Shielded Twisted Pair Cable, Fiber Optics.
- Patch Cords: UTP Patch Cords, Fiber Patch Cords. Cable Manager, Cable Tag, Cable Stripper, Scissors.
- Testing Tools: LAN Testers, Application Tools- IP Scanner, Test Network Speed, Bandwidth Monitor 2 Lite Edition, Netbrute Scanner.

UNIT-III

Establishing Lan Connection:

- Patching of LAN Cable, Strip Cable End, Untwist Wire Ends.
- Arrange Wires, Trim Wires to Size, Attach Connector, Check, Crimp, Test.
- Setting Up of LAN Using UTP Cables, Switch, LAN Tester, Switch Rack, Patch Panel, Cable Manager, Cable Tagging, I/O Box, I/O Face Plate, Patch Cords, Connectors, Terminals.

Networking Commands:

	• ping, ipconfig, fstat, iproute, route, netstat, ipconfig/release, ipconfig/renew. **Unit-IV** Internet Connectivity: • Different Types of Internet Connection (Wireless, Mobile, Hotspot, Dial-Up, Broadband, DSL, Cable, Satellite, ISDN.) Wi-Fi Connectivity: Basic Configuration of Wi-Fi Device or Access Point, IP Address Configuration, DNS Address, Security and Password, Admin Login and Password, MAC Filtering and Monitoring.
11	**Suggested Resources for Reading** 1. Microsoft Windows Networking Essentials 1st Edition, Darril Gibson, Sybex. 2. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education. 3. Cisco Networking Essentials, Troy Mcmillan, Sybex. 4. Comdex Hardware and Networking Course Kit, Vikas Gupta, 1st Edition, Dreamtech Press. 5. Networking: The Complete Reference 1st Edition, Mcgraw Hill Education. 6. Wireless Communication: Principle and Practices, Theodore S. Rappaport, Pearson Publication. **Note**: Latest edition of text books may be used.

Semester III

<table>
<tr><th colspan="2">Course Title</th><th>Basics of Routing and Switching</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-301</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
iv. The students are required to attempt one long question of 16 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, it enables the students to enhance career development skill in the field of networking.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Identify different types of routers, describe their functions, modes of operation and configure basic router settings.
CO2. Differentiate between static and dynamic routing, configure static routes.
CO3. Understand and implement dynamic routing protocols and verify their configurations.
CO4. Understand Switch Functions and its configuration.
CO5. Configure and Manage VLANs.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.</td></tr>
</table>

10	**Course Content** **UNIT-I** Router: Introduction, Components of routers, Types of Routers: Broadband Routers, Wireless Routers, Edge Router, Subscriber Edge Router, Inter-provider Border Router, Core Router, Wired and Wireless Routers. Functions of routers, Router modes, Configuration of Router: names, passwords, password encryption, interfaces name, creating a login banner, saving configuration and erasing configuration. **UNIT - II** Routing: Introduction, Types of routing: static and dynamic routing, Static routing: Configuring a static route, permanent keyword, static routes, and administrative distance, configuring default route. Dynamic routing: Dynamic Routing Protocols: RIPv1, RIPv2, IGRP, EIGRP and OSPF. Delivery Semantics: unicast, anycast, multicast, broadcast, geocast. Configuring and Verifying RIP, IGRP, OSPF, EIGRP. **UNIT-III** Switch: Introduction, roles of switches in network, types of switches: managed and unmanaged switch. Working of switch in different layers. Configuration of Switch: Command Mode, Setting Host Names, Setting Passwords, Setting IP Addresses and Default Gateways, Setting Operational Speed, Switch Port Security, Sticky MAC Addresses. **UNIT-IV** Switching: Introduction, Switching Services, Bridging vs LAN Switching, Switch Functions at Layer2, Definition of Spanning Tree Protocol (STP), Operations of STP, STP Port States, Definition of VLAN, Typing of VLAN: Static and Dynamic, VLAN Identification Methods, VLAN Trunking Protocol (VTP), VTP Modes of Operation, Routing between VLANs, Configuring Inter-VLAN Routing.
11	**Suggested Resources for Reading** 1. Lacoste, CCNP Routing and Switching TSHOOT 300 - 135: Official Cert Guide, Pearson India 2. Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd 3. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education 4. Vikas Gupta, Comdex Hardware and Networking Course Kit, 1st Edition, Dreamtech Press. 5. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media. 6. Networking: The Complete Reference 1st Edition, Mcgraw Hill Education

<table>
<tr><th colspan="2">Course Title</th><th>Programming Using C Language</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-302</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practicals
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>The objective of this course is to make the student understand programming language concepts, mainly control structures, reading a set of data, stepwise refinement, function and arrays. After completion of this course, the student is expected to analyze the real-life problem and write programs in ‘C’ language to solve problems. The main emphasis of the course is on problem solving aspect.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Understand the programming concepts.
CO2. Learn the C programming in practical usage.
CO3. Solve programming problems using C programming language.
CO4. Gain the in-depth knowledge of strings, arrays, functions and file handling etc.
CO5. Develop applications based on programming concepts.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.</td></tr>
</table>

10	**Course Content** **UNIT-I** Programming Process: Steps in developing a program, Algorithm development, Flowchart, Testing and Debugging. Fundamentals of C Languages: History of C, Character Set, Identifiers, Keywords, Constants, Types of C Constants, Variables, Data Types. Operators and Expressions: C Instructions, Arithmetic operators, Relational operators, Logical operators, Assignment Operators, Hierarchy of Operations, Structure of a C program, Compilation and Execution. **UNIT-II** Decision Control Structure: IF-statement, IF-Else and Nested IF-Else, The else if Clause. Loop Control Structure: While and do-while, for loop and Nested for loop, Case Control Structure: Decision using switch, goto statement. Functions: Library functions and user defined functions, Global and Local variables, Function Declaration, Calling and definition of function, Methods of parameter passing to functions, recursion. **UNIT-III** Arrays: Introduction, Array declaration, accessing values in an array, initializing values in an array, Single and Two-Dimensional Arrays, initializing a 2-Dimensional Array, Memory Map of a 2-Dimensional Array, Arrays of characters, Insertion and deletion operations. Pointers: Pointer declaration, Address operator "&", Indirection operator "*", simple implementation of pointers. **UNIT-IV** String Manipulation in C: Declaring and Initializing string variables, Reading and writing strings, String functions (strlen(), strcpy(), strcmp(), strcat()). Structures and Unions: Declaration of structures, Structure Initialization, Accessing structure members, Union. Files in C: Introduction, Opening and Closing files, Basic I/O operation on files.
11	**Suggested Resources for Reading** 1. Yashavant P. Kanetkar: Let us C, BPB Publications, New Delhi. 2. Salaria, R.S.: Test Your Skills in C, Salaria Publications, New Delhi. 3. C. Balaguruswami: Programming with C Language, Tata McGraw Hill, New Delhi. 4. Byron S. Gottfried: Programming in C, McGraw Hills Publishers, New York. 5. M.T. Somashekara: Programming in C, Prentice Hall of India. **Note**: Latest edition of text books may be used.

BHN-302P (Practical Based on BHN-302)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

<table>
<tr><th colspan="2">Course Title</th><th>Routing Configuration</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-303</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>120 hours of practicals
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>-</td><td>-</td><td>80</td><td>20</td></tr>
</table>
The exam of this course will be of 3hours duration.
i. There will be no objective type questions.
ii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iii. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
Instructions for paper setters and candidates:
i. Examiner will set total of four questions covering the whole syllabus.
ii. Student will attempt any three questions.
iii. Each question carries 10 marks.
iv. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>Routing technology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Describe basic routing concepts, configure and verify basic router settings using CLI.
CO2. Differentiate between static and dynamic routing, link state and distance vector protocols, and analyze key routing concepts.
CO3. Configure and verify inter-VLAN routing.
CO4. Test network connectivity and troubleshoot configurations.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content

UNIT-I
Functionality of Router and Switch: Describe basic routing concepts: packet forwarding, router lookup process, Process Switching/Fast Switching/CEF.
Configure and verify utilizing the CLI to set basic Router configuration: hostname, local user and password, enable secret password, console and VTY logins, exec-timeout, service password encryption, Interface IP Address, loopback, banner, motd, copy run start.

UNIT-II
Configure and verify operation status of a device interface: Serial, Ethernet, Verify router configuration</td></tr>
</table>

	and network connectivity using, ping, extended, traceroute, telnet, SSH, shcdpneighbors**.** The boot process of Cisco IOS routers, Configure and verify operation status of a device interface, both serial and Ethernet, Verify router configuration and network connectivity. IP Routing Technologies: Configure and verify routing configuration for a static or default route given specific routing requirements. **UNIT-III** Introduction to routing and routing protocols, Static versus Dynamic Routing, 4.6.b Link state versus Distance Vector, next hop, IP routing table, Passive Interfaces and its working, Admin Distance, split horizon, metric. Configure and Verify OSPF: Benefit of single area, Configure OSPv2, Configure OSPv3, Router ID, Passive Interface, Multi-area OSPF, LSA types and purpose. **Unit-IV** Configure and verify inter VLAN routing (Router on a stick), sub interfaces, upstream routing, encapsulation, Configure SVI interfaces, Manage Cisco IOS Files, Boot Preferences, Cisco IOS Images (15), Licensing, Show license, Change license, Configure and verify EIGRP (single AS), Feasible Distance/Feasible Successors/Administrative distance, Feasibility condition, Metric composition, Router ID, auto summary, Path Selection, Load Balancing, Unequal, Equal.
11	**Suggested Resources for Reading** 1. Lacoste, CCNP Routing and Switching TSHOOT 300-135: Official Cert Guide, Pearson India 2. Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd 3. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education 4. Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons 5. Crypto and Network Security, 3rd Edition, Mcgraw Hill Education 6. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media **Note**: Latest edition of text books may be used.

Semester IV

<table>
<tr><th colspan="2">Course Title</th><th>Digital Electronics and Computer Organisation</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-401</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>80</td><td>20</td><td>-</td><td>-</td></tr>
</table>
Theory Paper:
(4 X 4 marks) = 16 marks
(4 X 16 marks) = 64 marks
i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 4 marks each.
iv. The students are required to attempt one long question of 16 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course is an introduction to the basic principles of digital electronics and computer organisation. At the conclusion of this course, the student will be able to quantitatively identify the fundamentals of computers, including number systems, logic gates, logic, arithmetic subsystems, and integrated circuits. The student will be able to analyze and design simple logic circuits and draw logic diagrams.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Understand significance of number systems and its conversions.
CO2. Illustrate knowledge on design of various combinational circuits.
CO3. Illustrate the concept of sequential logic design, analyze the operation of flip-flops, registers, and counters
CO4. Discuss the basic structure and organization of computers.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.</td></tr>
</table>

10	**Course Content** **UNIT-I** Introduction to Digital Electronics: Digital vs Analog Systems, Analog vs digital signals, advantages of digital systems, Number Systems (Binary, Decimal, Octal, Hexadecimal) and Conversion between different number systems and Binary Arithmetic (Binary addition, subtraction). Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR), Truth tables and symbolic representations. Introduction to Computer Systems: Basic Computer Architecture, Stored program concept and Von Neumann architecture, Basic Operations in a Computer (CPU operations: Fetch-Decode-Execute cycle), Arithmetic and Logic Unit (ALU) and Control Unit (CU). **UNIT - II** Combinational Circuits: Half adder, Full adder, Subtractors, Multiplexers, Demultiplexers, Encoders, Decoders. CPU Design: Components of the CPU: ALU, Registers and Types of registers (Accumulator, Program Counter, Instruction Register), Internal communication: Buses, data lines, address lines, Basic understanding of the Instruction Set Architecture (ISA), addressing modes, instruction pipelining**.** **UNIT-III** Instruction Types: Data transfer, Arithmetic, Logic, Branching, and Control instructions. Example of a basic instruction set for simple processors. Sequential Circuit: Flip-Flops (SR, D Flip-Flop, T Flip-Flop, JK Flip-Flop) and their truth tables, race around condition, Master Slave flip flop, Shift registers: SIPO, PISO, PIPO, SIPO, Bi-directional; Counters: ripple & synchronous counters – up / down. **UNIT-IV** Memory Organization and Architecture: Memory Hierarchy, Cache memory (levels of cache, hit/miss rate), RAM, ROM, Virtual Memory. Memory Access and Optimization: Memory Access Techniques (Direct Memory Access (DMA), Memory-Mapped I/O), Cache Misses and Optimization. Processor Performance and Optimization: Processor Performance Metrics, Performance Optimizations.
11	**Suggested Resources for Reading** 1. M. Morris Mano. Computer System Architecture, Prentice Hall International 3 rd ed., (1993). 2. P. Pal Choudhar. Computer Organisation and Design, Prentice Hall of India, (1994). 3. B. Govindarajalu, IBM-PC and Clones – Hardware Troubleshooting and Maintenance, Tata – McGraw Hill, (1994).

<table>
<tr><th colspan="2">Course Title</th><th>Network Administration</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-402</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practicals
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>The goal of this course is to train students to an exceptionally high standards in network planning, implementation, administration and support.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Perform IOS Software and Configuration Management tasks.
CO2. Understand and Implement Password Recovery Procedures.
CO3. Apply Basic Troubleshooting Techniques and Enhance Network Security and Performance.
CO4. Utilize Security Device Manager (SDM) for Network Configuration.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content
UNIT-I
Backing UP and Restoring IOS Software and Configuration: Boot system commands, IOS file system, backing up configuration to a TFTP server, restoring configuration to a TFTP server, backing up the IOS software to a TFTP server, restoring/upgrading the IOS software from a TFTP server, restoring IOS</td></tr>
</table>

	software from ROM Monitor mode using Xmodem. **UNIT-II** Password-Recovery Procedures and configuration Register: The configuration register: A visual representation, bit means, the boot field, console terminal baud rate setting. Console line speed: CLI & ROM Monitor mode, password recovery procedures for router, password recovery procedures for switches. **UNIT-III** Basic troubleshooting: viewing routing table, determining the gateway of last resort, last routing update, OSI layer 3 testing, OSI layer 7 testing, interpreting the show interface command, using CDP to troubleshooting, show controllers command debug commands, time stamps, Operating system IP verification commands, the ip http server command, netstat command. **UNIT-IV** Security Device Manager: SDM connecting with CLI and GUI, SDM Express Wizard with no CLI preconfiguration, resetting the router to factory defaults, SDM user interfaces, configuration routing using SDM, SDM monitor mode, SDM to configuration a router as a DHCP server & client, SDM configuration NAT/PAT.
11	**Suggested Resources for Reading** 1. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education 2. Cisco Networking Essentials, Author: McMillan, troy author; John Wiley & Sons 3. Vikas Gupta, Comdex Hardware and Networking Course Kit, 1st Edition, Dreamtech Press 4. Crypto and Network Security, 3rd Edition, Mcgraw Hill Education 5. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media 6. Networking: The Complete Reference 1st Edition, Mcgraw Hill Education **Note**: Latest edition of text books may be used.

BHN-402P (Practical Based on BHN-402)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

<table>
<tr><th colspan="2">Course Title</th><th>DOS and Windows Operating System</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-403</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practicals
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
Theory Paper:
(4X 2 marks) = 8 marks
(4X8 marks) = 32 marks

i. The exam of this course will be of 3 hours duration.
ii. There will be no objective type questions.
iii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iv. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

Instructions for paper setters and candidates:
i. The syllabus of this paper has been divided into FOUR units.
ii. All questions carry equal marks
iii. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. In Question no. 1, students are required to attempt 4 out of 6 short questions of 2 marks each.
iv. The students are required to attempt one long question of 8 marks from each unit.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>This course enables students to understand the concepts of Operating System. After the completion of this paper, student will be able to manage various processes, files and folders on the disk and desktop with effective outcome.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Demonstrate proficiency in DOS.
CO2. Understanding Windows operating systems and performing basic program management tasks.
CO3. Configure system properties, manage devices through the device manager, and use troubleshooting tools.
CO4. Utilize various tools for maintaining and optimizing storage devices, and managing folders and files.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
</table>

10	**Course Content** **UNIT-I** DOS: Versions of DOS: Booting sequence; Warm and Cold reboot; Concept of File and directory, Redirecting command input and output pipes, Wildcard characters, Types of DOS commands: Internal and External; Internal Commands: DIR, MD, CD , CLS, COPY, DATE, DEL, PATH, PROMPT, REN, RD, TIME, TYPE, VER, VOL; External Commands: XCOPY, ATTRIB, BACKUP, RESTORE, FIND, SYS , FORMAT, CHKDSK, DISKCOPY, LABEL, MOVE, TREE, DELTREE, DEFRAG, SCA NDISK, UNDELETE. Batch Files: Introduction to simple batch files; Introduction to CONFIG.SYS and AUTOEXEC.BAT files. **UNIT-II** Windows Operating System: features, history, hardware requirements, and installation of Windows Operating System. Graphics interface: Benefits, Screen attributes: icons and bars, Accessories of the Windows program. Program management: Creating, saving and editing documents, Concurrent execution of programs, sharing data between programs. Folder and file management: Working with files, Naming files, Navigate to Folders with Windows Explorer, Copying and moving files, deleting files, managing folders, Creating, Viewing, Expanding and collapsing, Backing up and restoring files. **UNIT-III** Components of Windows: format of a window, moving windows, resizing windows, minimizing and maximizing windows. Control panel: Customizing screens, Screen colours, Patterns, Spacing icons, selecting time/date, setting the Sound. Creating Short cuts, Notepad, System restore. Customizing printing, changing the print queue, configuring the printer, Adding printers. Working with fonts: changing, removing, adding, customizing mouse and keyboard use. **UNIT-IV** System properties and the device manager Management tools, DOS sessions, Explorer, Memory configuration, Safe mode Install and uninstall applications, Setup/troubleshooting issues. Maintaining and optimizing disks: Disk Cleanup, Disk defragmenter, Compressing and uncompressing folders and files. Encrypting and decrypting folders and files.
11	**Suggested Resources for Reading** 1. Jamsa, Kris A., Rescued by Windows 3.1, Galgotia. 2. Basandra, S.K., Computers Today, Galgotia. 3. Mathur Rajiv, Learning Word 6 for Windows Step-by-Step, Galgotia. 4. Mathur Rajiv, Learning Excel 5 for Windows Step-by-Step, Galgotia. **Note**: Latest edition of text books may be used.
	BHN-403P (Practical Based on BHN-403) Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation. 1. Examiner will set total of four questions covering the whole syllabus. 2. Student will attempt any three questions. 3. Each question carries 5 marks. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

<table>
<tr><th colspan="2">Course Title</th><th>Switching Configuration</th></tr>
<tr><td>1</td><td>Course Code</td><td>BHN-404</td></tr>
<tr><td>2</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>3</td><td>Course Duration</td><td>120 hours of practicals
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>4</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>5</td><td>Max. Marks, Time and Instructions</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr>
<tr><td>-</td><td>-</td><td>80</td><td>20</td></tr>
</table>
The exam of this course will be of 3hours duration.
i. There will be no objective type questions.
ii. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
iii. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
Instructions for paper setters and candidates:
i. Examiner will set total of four questions covering the whole syllabus.
ii. Student will attempt any three questions.
iii. Each question carries 10 marks.
iv. Practical file carries 10 marks and 40 marks for practical work and viva-voce.
</td></tr>
<tr><td>6</td><td>Job Role</td><td>Network Support Engineer</td></tr>
<tr><td>7</td><td>Course Objectives</td><td>Switching technology provides an integrated and comprehensive coverage of networking topics from fundamentals to advanced application and services, while providing opportunities for hands on practical experiences and career skill development.</td></tr>
<tr><td>8</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
CO1. Gain proficiency in configuring and verifying DHCP, ACLs, and NAT for network operations, addressing various network requirements and enhancing security.
CO2. Configure and verify NTP, Syslog, and SNMP to ensure synchronized time, event logging, and effective network monitoring.
CO3. Configure and verify switch operations and understand switching methods and concepts.
CO4. Configure enhanced switching technologies, port security features, secure unused ports and optimize VLAN configurations to strengthen network security.</td></tr>
<tr><td>9</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>10</td><td colspan="2">Course Content

UNIT-I
IP Services: Configure and verify DHCP (IOS Router), Configuring router interfaces to use DHCP, DHCP options (basic overview and functionality), Excluded addresses, Lease time, Describe the types, features, and applications of ACLs, standard (editing and sequence numbers) extended, named,</td></tr>
</table>

	numbered, Log option. Configure and verify ACLs in a network environment, named, numbered, Log option, Identify the basic operation of NAT, purpose, pool, static, 1 to 1, overloading, source addressing, one way NAT, Configure and verify NAT for given network requirements. **UNIT-II** Configure and verify NTP as a client. Recognize High availability (FHRP), VRRP, HSRP, GLBP, and Configure and verify syslog, Utilize Syslog Output, Describe SNMP v2 and v3. LAN Switching Technologies: Determine the technology and media access control method for Ethernet networks, Identify basic switching concepts and the operation of Cisco switches: Collision Domains, Broadcast Domains, Ways to switch, Store, Forward, Cut through, CAM Table. **UNIT-III** Configure and verify initial switch configuration including remote access management: Hostname, management IP address, IP default-gateway, local user and password, enable secret password Console and VTY logins, exec-timeout service password encryption, copy run start, verify network status and switch operation using basic utilities such as: ping, telnet, SSH. Describe how VLANs create logically separate networks and the need for routing between them, network segmentation and basic traffic management concepts, Configure and verify VLANs. **Unit-IV** Configure and verify trunking on Cisco switches: DTP (topic), auto-negotiation, Identify enhanced switching technologies: RSTP, PVSTP, Ether channels, Configure and verify PVSTP operation: Describe root bridge election, Spanning tree mode. Configure and verify Switch Port Security: Sticky mac, MAC address limitation, static/dynamic, violation modes, err disable, shutdown, protect restrict, shutdown unused ports, err disable recovery, assign unused ports in unused VLANs, putting Native VLAN to other than VLAN 1.
11	**Suggested Resources for Reading** 1. Lacoste, CCNP Routing and Switching TSHOOT 300 - 135: Official Cert Guide, Pearson India 2. Todd Lammleccna, Routing and Switching Study Guide, Wiley India Pvt Ltd 3. CCNA Cisco Certified Network Associate Study Guide 3rd Edition, Mcgraw Hill Education 4. Mcmillan,Cisco Networking Essentials, Author: John Wiley & Sons 5. Crypto and Network Security, 3rd Edition, Mcgraw Hill Education 6. Cisco a Beginner's Guide, Fifth Edition, McGraw-Hill/Osborne Media **Note**: Latest edition of text books may be used.